

COMP3052.SEC Computer Security

Session 15-1 Crypto I

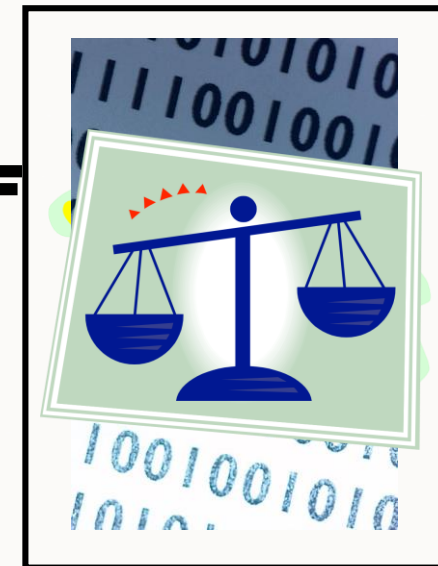


ACKNOWLEDGEMENTS

- Some of the materials we use this semester may come directly from previous teachers of this module, and other sources ...
- Thank you to (amongst others):
 - Michel Valstar, Milena Radenkovic, Mike Pound, Dave Towey ...

TOPICS COVERED

- Concepts of Cryptology
- The Mathematics of Cryptology
- Algorithms and Mechanisms
 - Integrity Checking
 - Digital Signatures
 - Encryption
- Assessment of Algorithms and Mechanisms



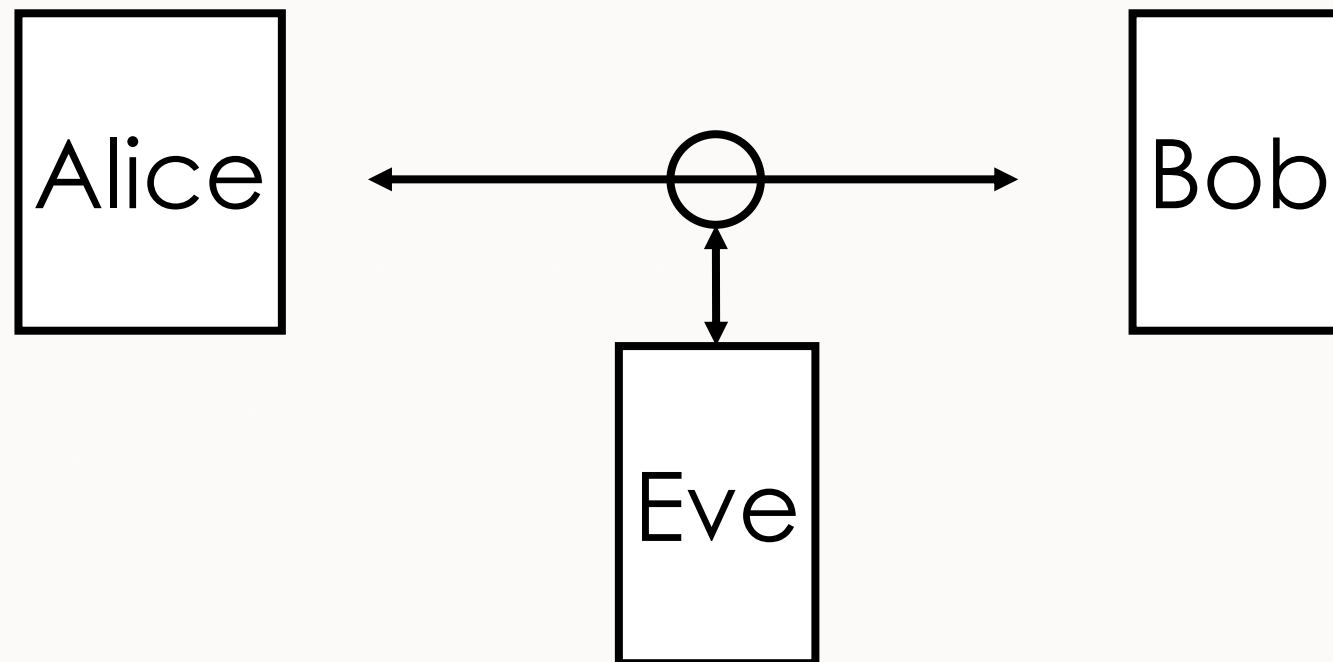
DEFINITIONS

- ***Cryptography***

- ***Cryptanalysis***

- ***Cryptology***

THE COMMUNICATION MODEL



ALL ABOUT EVE

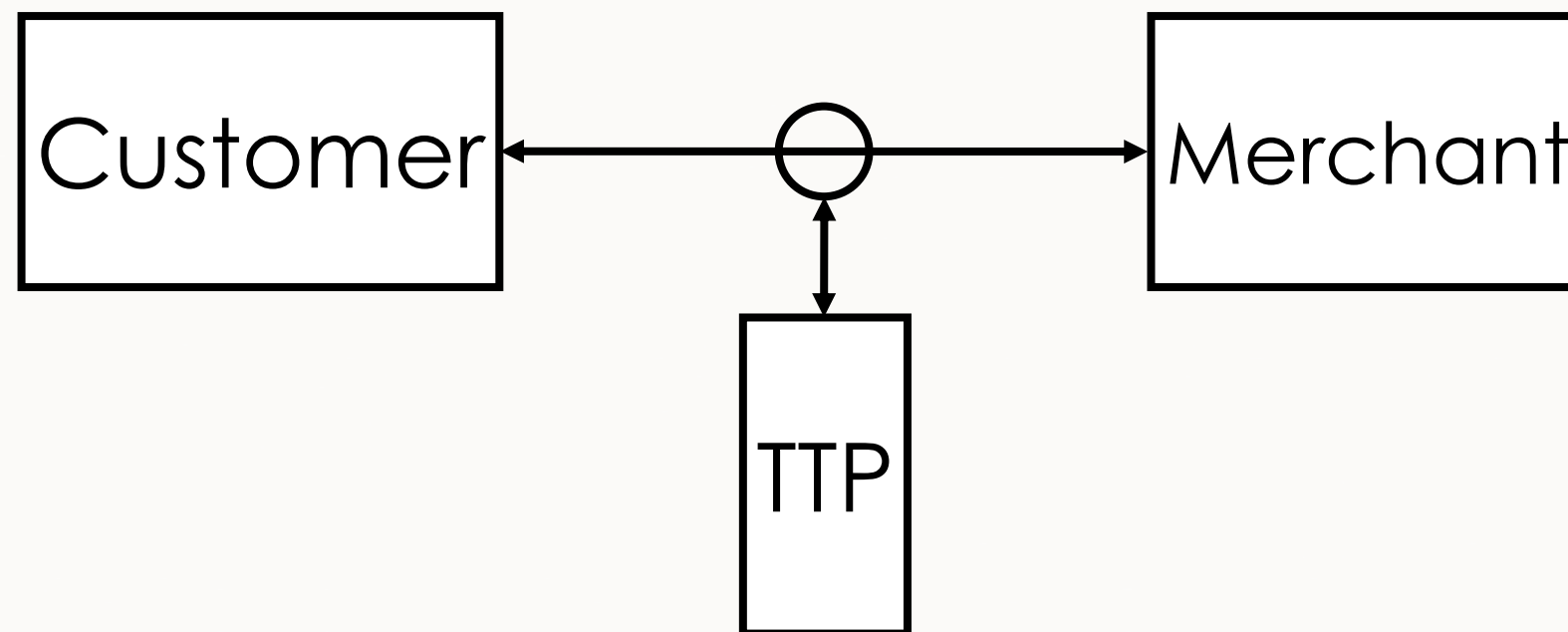
Objectives

- Read messages
- Edit Messages
- Write messages

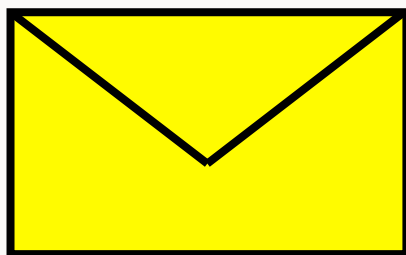
Countermeasures

- Encryption
- Integrity Checking
- Origin Checking

THE COMPUTER SECURITY MODEL



CRYPTOGRAPHY PARADIGMS

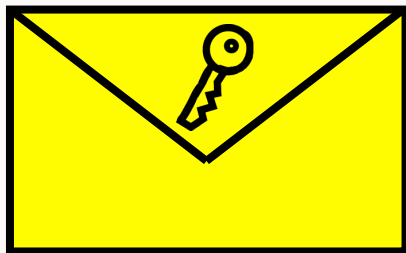


SECRET
PROCESS

ACTIVITY ...

- Please answer:
 - What happens if someone reversely engineers/sells your process?
 - Who verifies that your technique is strong enough?

THE NEW PARADIGM



Known
process



KEYS AND STRENGTH

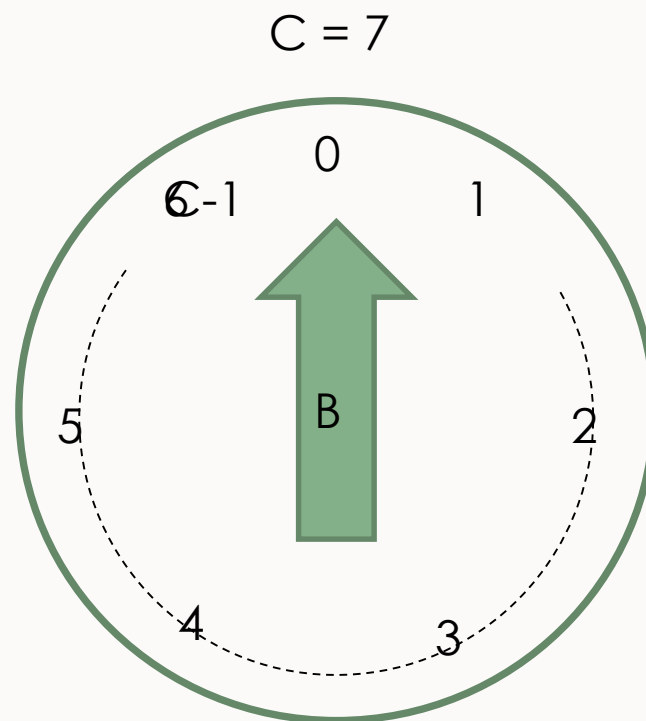
- “Security is as strong as its weakest link”
– Bruce Schneier
- Keys vary in strength
- Algorithms vary in strength
- “Brute Force” attacks

KEY MANAGEMENT

- Where do we generate keys?
- How do we generate keys?
- Where are keys stored?
- How do we transport keys?
- Where are keys used?
- How are keys revoked and replaced?

MODULUS MATHEMATICS

$$A = 00$$



MODULUS MATHEMATICS

$$A \equiv B \pmod{C}$$

PROPERTIES OF THE MODULUS

Let

$$a_1 \equiv b_1 \pmod{C}$$

$$a_2 \equiv b_2 \pmod{C}$$

Then

$$a_1 + a_2 \equiv (b_1 + b_2) \pmod{C}$$

$$a_1 a_2 \equiv (b_1 b_2) \pmod{C}$$

PROPERTIES OF THE MODULUS

- Constrained to integers so division is complex
- Let's define division as the opposite of multiplication
- If $f = (1/e) \bmod C$
 - then $(e*f) \bmod C = 1$
- If multiple prime solutions exist for $1/e$, the division is **undefined** like $X/0$ in regular mathematics

ACTIVITY ...

- Calculate the value of f :

$$f = (1/3) \text{ Mod } 7$$

ACTIVITY ...

- Calculate the value of f:

$$f = (1/3) \text{ Mod } 7$$

- $f = (1/3) \text{ Mod } 7$
- So $(3f) \text{ Mod } 7 = 1$
- Possible values of f:
 - $0, 0 \text{ Mod } 7 = 0$
 - $1, 3 \text{ Mod } 7 = 3$
 - $2, 6 \text{ Mod } 7 = 6$
 - $3, 9 \text{ Mod } 7 = 2$
 - $4, 12 \text{ Mod } 7 = 5$
 - **$5, 15 \text{ Mod } 7 = 1$**
 - $6, 18 \text{ Mod } 7 = 4$
- So $f = (1/3) \text{ Mod } 7 = 5$

UNDEFINED MODULAR DIVISION

- What about
 - $f = (5/5) \text{ Mod } 10$
 - $(5f) \text{ Mod } 10 = 5$

UNDEFINED MODULAR DIVISION

- What about
 - $f = (5/5) \bmod 10$
 - $(5f) \bmod 10 = 5$
- $0, 0 \bmod 10 = 0$
- $1, 5 \bmod 10 = 5$
- $2, 10 \bmod 10 = 0$
- $3, 15 \bmod 10 = 5$
- $4, 20 \bmod 10 = 0$
- Etc

Undefined – too many prime answers!
There are cases where there are no answers!

PRIMES AND MODULUS

- Let p be a prime number
- Let a be an integer

- iff $a \not\equiv 0 \pmod{p}$

There is always another integer d such that:

$$a * d = 1 \pmod{p}$$

FERMAT'S LITTLE THEOREM

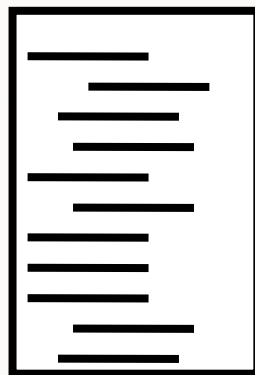
- iff $a \not\equiv 0 \pmod{p}$

$$a^{p-1} \equiv 1 \pmod{p}$$

- This yields a good way of testing if a number is prime
 - If you calculate $a^{p-1} \pmod{p}$ for a series of numbers if there are no 1's it probably isn't prime!

HASH FUNCTIONS

X



Any number of bits

HASH
FUNCTION

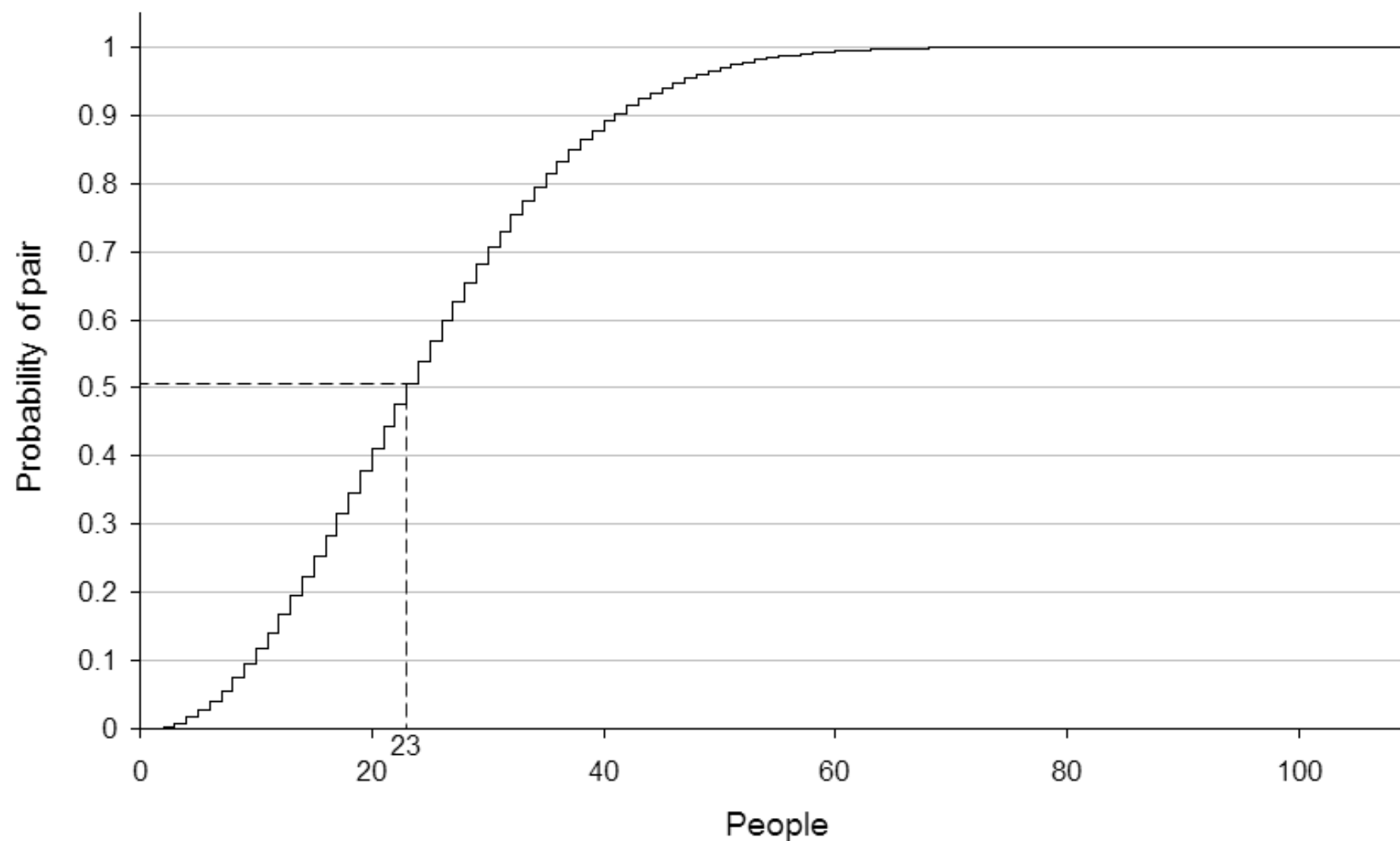
HASH FUNCTION PROPERTIES

- Compression
 - No matter how long the input is, the output has the same length
- Ease of computability
 - Given x , it should be easy to find $h(x)$
- Collision Avoidance
 - It should be “computationally infeasible” to find collisions

QUESTION/ACTIVITY ...

THE BIRTHDAY PARADOX

- How many of you share a birthday?



HASH FUNCTION PROPERTIES

- Given a hash function that produces N bit hashes
 - If you generate around $2^{N/2}$ random inputs, you are likely to find a collision

HASH FUNCTION PROPERTIES

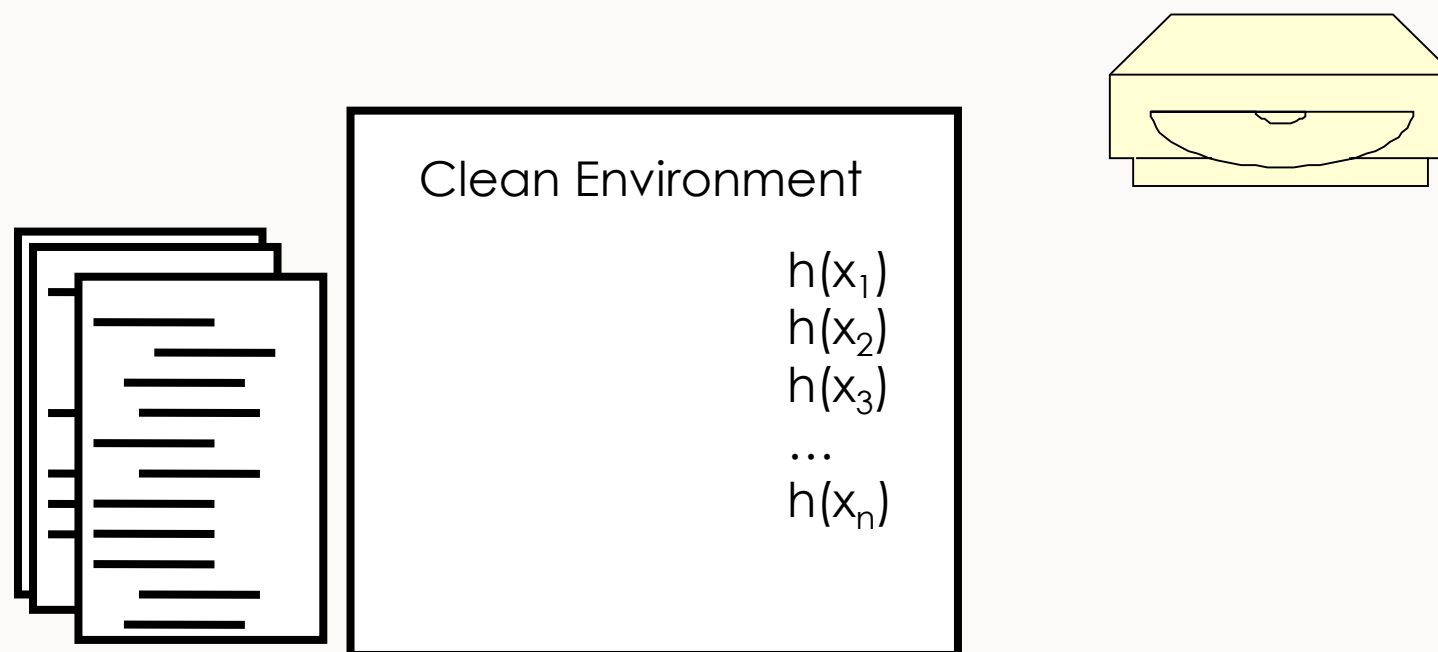
- Preimage Resistance
 - Given y , it should be “computationally infeasible” to find x to satisfy:
 - $h(x) = y$
 - Expected number of tries is 2^{n-1} for an n -bit hash
- Second Preimage Resistance
 - (weak collision resistance)
 - Given x and $h(x)$, it should be “computationally infeasible” to find x' to satisfy
 - $h(x) = h(x')$

HASH FUNCTION PROPERTIES

- Collision Resistance
 - (strong collision resistance)
 - It should be “computationally infeasible” to find any x and x' that satisfy:
 - $h(x) = h(x')$

MDC-2

- Modification Detection Codes 2
- A cryptographic hash function



SUMMARY

- What is cryptology?
- Communication/encryption paradigms
- Modulus Mathematics
- Hash functions
- MDCs

Read:

- Gollman: Chapter 14
- Anderson: Chapter 5